

Md. Sakib Hasan

Full-Stack Developer | AI Engineer

📍 Aftabnagar, Dhaka 📞 +880 1790 639339 ✉️ sakibmunshi013@gmail.com
in md-sakib-hasnan-356304272 🌐 Hasan-Sakib 📁 Portfolio

Professional Summary

Results-driven AI Engineer and Full-Stack Developer with a strong foundation in building scalable web applications and end-to-end intelligent systems. Proficient in designing robust backend architectures and responsive frontends while integrating machine learning models, with a specialization in computer vision, into production-ready pipelines. Published researcher with hands-on experience applying deep learning frameworks such as PyTorch and TensorFlow to solve real-world problems. Adept at bridging the gap between software engineering and artificial intelligence, delivering solutions that are both technically rigorous and business-impactful. Passionate about leveraging emerging AI technologies to build the next generation of intelligent, data-driven applications.

Education

Bachelor's Degree in Computer Science & Engineering Major in Data Science

East West University, Dhaka
CGPA: 3.46 / 4.00

January 2022 – December 2025

Higher Secondary Certificate (HSC), Science

Bangladesh Navy College, Dhaka
GPA: 5.00 / 5.00

2018 – 2020

Projects

Vision-Based Self-Supervised Malware Detection with Web Application (Final year project)

Developed an AI-based malware detection system using computer vision techniques by converting executable files into image representations for classification. Self-supervised learning was applied to reduce dependency on labeled malware datasets while improving feature learning. A web-based application was built that allows users to upload files and receive real-time malware predictions. The trained models were integrated with backend APIs to enable efficient deployment and system functionality.

LangGraph AI Agent Chatbot



- Built a full-stack conversational AI system using a LangGraph ReAct agent, FastAPI backend, and Streamlit UI; engineered runtime multi-provider LLM routing across Groq (Llama 3.3-70B, Llama 3.1-8B) and OpenAI GPT-4o-mini — switchable at chat time without service restarts
- Integrated Tavily web search as an agentic tool with server-side session state management.
- Containerised frontend and backend as two decoupled Docker images orchestrated with Docker Compose, ensuring consistent, cloud-ready deployment

Abiyan Telestore – Production-Ready E-Commerce Platform



- Architected a fully containerised, three-service e-commerce platform (Customer Storefront, Admin Dashboard, FastAPI backend) orchestrated via Docker Compose with an Nginx reverse proxy and automated GitHub Actions CI/CD pipeline
- Engineered a dual-token JWT auth system, TypeScript client auto-generated from FastAPI's OpenAPI spec to eliminate API drift, and Cloudinary CDN for optimised image delivery
- Delivered an optimistic, animation-rich React 18 frontend with TanStack Query for server-state caching, Zustand for client state, GSAP micro-interactions, and transactional email via SMTP

12 Python Projects — Algorithms, AI, Games & Image Processing



- Built a curated collection of 12 self-contained Python projects spanning algorithms (Binary Search, Sudoku Solver with backtracking), AI/NLP (Markov Chain Text Composer, Minimax-based Tic Tac Toe AI), and image processing (Photo Manipulation with NumPy pixel operations)
- Reinforced core Python concepts including recursion, OOP, probability, graph theory, and 2D array manipulation through hands-on, project-driven implementations

Research Publication

Freshwater Fish Species Classification Using Self-Supervised Learning

[Paper Link]

Published at the 2025 28th International Conference on Computer and Information Technology (ICCIT) – IEEE

- Developed a self-supervised learning framework for freshwater fish image classification, training on unlabeled data to learn robust visual representations fine-tuned for downstream tasks
- Outperformed traditional supervised baselines in low-labeled-data scenarios, demonstrating strong generalization with minimal annotation overhead

Technical Skills

AI / ML Engineering

Core Concepts: Machine Learning, Deep Learning, Representation Learning

Computer Vision: CNNs, Image Classification, Object Detection, Feature Extraction

Generative AI & LLMs: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Prompt Engineering, Agentic AI Systems

AI Frameworks & Orchestration: PyTorch, TensorFlow, Scikit-learn, OpenCV, LangChain, LangGraph

Data & Analysis: Pandas, NumPy, Matplotlib, Jupyter Notebook, Google Colab

Full-Stack Development

Languages: Python, JavaScript, Typescript, Java, C, SQL

Frontend: React.js, HTML5, CSS3, Tailwind CSS

Backend: Node.js, FastAPI, REST APIs, JWT Authentication, Object-Oriented Programming (OOP)

Databases & DevOps: PostgreSQL, MySQL, Git, GitHub, Docker, CI/CD pipeline

Research Interests

- Agentic AI Systems, LLM Orchestration, and Multi-Agent Frameworks
- Retrieval-Augmented Generation (RAG) and Knowledge-Grounded Language Models
- Self-Supervised and Representation Learning for Vision and NLP
- Computer Vision, Multimodal Learning, and AI-Driven Image Analysis
- Scalable AI System Design and Production ML Engineering

Languages

- **Bengali** — Native
- **English** — Professional Proficiency (Reading, Writing, Speaking)